Large-Scale Retail Data Analytics

Strategic Business Intelligence Dashboard (6.7 million Records)

Project Overview

This project demonstrates advanced proficiency in **Excel Power Tools (Power Query & Power Pivot)** to manage and visualize massive datasets that exceed standard Excel limitations. By using a relational data model, I transformed nearly 7 million rows of raw CSV data into an interactive executive dashboard.

Technical Proficiencies Demonstrated

- **Big Data Management**: Processing 6,758,125 transactions within the Excel Data Model.
- **ETL Engineering**: Advanced data cleaning and multi-table merging using Power Query.
- **Relational Modelling**: Designing a Star Schema to link Sales, Cities, and Categories.
- **DAX (Data Analysis Expressions)**: Creating calculated measures for financial reporting.
- **UI/UX Dashboard Design**: Building interactive slicers and timelines for user-driven exploration.

Tools Used: Microsoft Excel, Power Query, Power Pivot, DAX

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1. Introduction

1.1 Project Context

This project involves the end-to-end processing and analysis of a large-scale retail dataset comprising approximately **6,758,125 transaction records** from the 2018 fiscal year. The data captures a wide array of retail activities, including product categories, geographical sales distribution, and customer transaction details. As businesses increasingly rely on data-driven strategies, the ability to transform millions of raw data points into actionable insights is a vital competency in the field of Business Information Technology.

1.2 Problem Statement

A significant challenge in modern data analytics is the limitation of standard spreadsheet software. Microsoft Excel has a traditional worksheet limit of 1,048,576 rows, making it impossible to analyse a dataset of **6.7 million rows** using standard grid methods. This project addresses the need for scalable architecture by utilizing the Excel **Data Model** and **Power Pivot** to bypass these physical constraints while maintaining system performance and data integrity.

1.3 Project Objectives

The primary objective of this study was to develop a professional-grade Business Intelligence (BI) dashboard that remains responsive despite high data volume. Specific goals included:

- **Data Integration**: Merging multiple CSV source files (Sales, Products, Customers, Cities, and Categories) into a unified relational stream using **Power Query**.
- **Architectural Design**: Implementing a **Star Schema** to establish efficient relationships between the central "Sales" fact table and surrounding dimension tables.
- **Advanced Analytics**: Utilizing **DAX (Data Analysis Expressions)** to calculate total line revenue and track performance metrics.
- **Strategic Visualization**: Creating an interactive interface with **Slicers** and **Timelines** to allow stakeholders to drill down into specific regions or time periods instantly.

2. Executive Summary

This project involved the end-to-end processing of a massive retail dataset comprising over **6.7 million** transactions from the year 2018. By utilizing advanced Excel features like **Power Query** and the **Data Model**, the system successfully bypassed traditional worksheet row limits to provide real-time insights into revenue performance.

Key Business Insight:

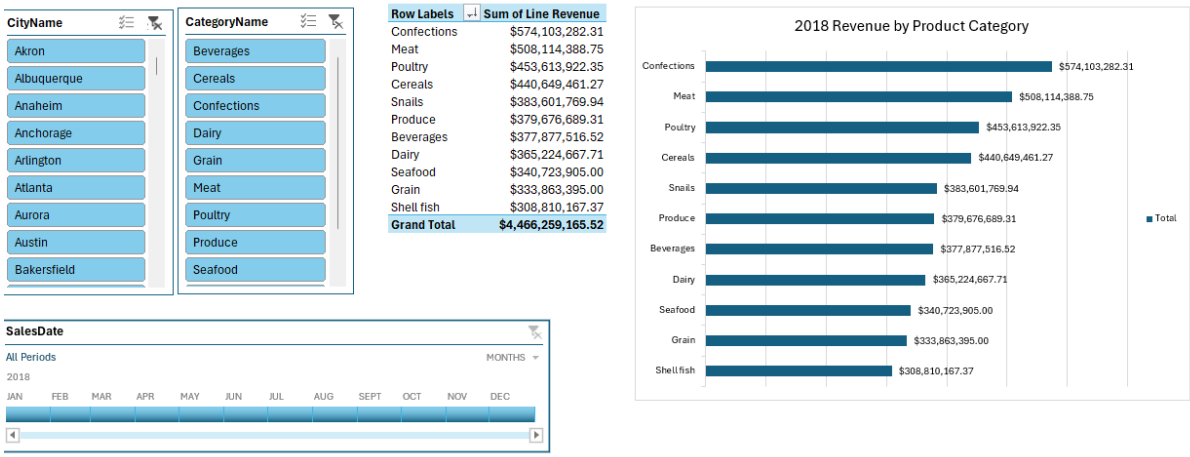
- **Top Performer:** The **Confections** category emerged as the primary revenue driver, generating over **$574 million** in total sales.
- **Scalability:** The architecture supports seamless filtering by **City**, **Category** and **Time**, despite the high data volume.

3. Technical Architecture (The "How")

- **Data Integration (Power Query):** Used to clean and merge multiple source files (Sales, Products, Customers, Cities, and Categories) into a unified stream.
- **Bypassing Grid Limits:** Because the dataset exceeded 1.04 million rows, the data was loaded as a **Connection Only** and added directly to the **Excel Data Model (Power Pivot)**.
- **Relational Modelling:** Established primary and foreign key relationships (e.g., CityID, CategoryID) to ensure data integrity across disparate tables.
- **DAX Calculations:** Created a custom **Line Revenue** measure using the logic =[Quantity] *[products.Price]

4. Dashboard Features

- **Dynamic Slicers:** Allows stakeholders to drill down into specific geographic regions (Cities) or products (Categories) instantly.
- **Timeline Analysis:** A dedicated time-series filter set to a **Monthly** allowing for the observation of seasonal trends within the 2018 fiscal year.
- **Visual Representation:** A sorted Pivot Chart that provides an immediate ranking of category performance formatted with consistent currency precision.



5. Technical Challenges & Solutions

- Dataset too large for Excel - Loaded data into the **Power Pivot Data Model** instead of the worksheet grid.
- Data Type Mismatches - Standardized CategoryID to **Whole Number** to enable Summing.
- File Access Conflicts**:** Resolved [DataSource.Error] by ensuring all source CSV files were closed during the data refresh process, allowing Power Query to establish an exclusive lock for loading.

6. Conclusion

The successful implementation of this analytics dashboard proves that Excel, when combined with Power Pivot and Power Query, is a viable tool for "Big Data" business intelligence. By managing over **6.7 million rows** of data, this project demonstrates the efficiency of the **Star Schema** in maintaining data integrity and system performance. The analysis clearly identifies **Confections** as the high-growth driver for the business in 2018, while the interactive slicers provide the detail needed for regional performance reviews.

7. Strategic Recommendations

Based on the visual insights generated by the dashboard, the following business actions are recommended:

- **Inventory Optimization**: Given that **Confections** is the top revenue generator ($574M+), the supply chain should prioritize stock availability for this category to prevent stockouts and lost revenue.
- **Regional Marketing Campaigns**: Using the **CityName** slicer, management should identify cities where Confections underperforms compared to the global average and launch targeted promotional campaigns in those specific areas.

- **Seasonal Resource Allocation**: By analysing the **Timeline**, the business should align staffing and logistics with peak sales months identified in the 2018 data to handle increased transaction volumes efficiently.
- **Product Diversification**: Investigate the **Shellfish** and **Grain** categories, which are currently at the bottom of the revenue chart to determine if they should be rebranded discounted or phased out in favour of higher-margin items.

8. Reflection on BBIT Skills Applied

This project served as a practical application of core BBIT competencies, including:

- **Database Management**: Applying relational algebra to link fact and dimension tables.
- **Business Intelligence**: Converting raw CSV data into actionable executive insights.
- **Systems Integration**: Connecting disparate data sources into a single, cohesive reporting engine.